



Module Name: MACHINE LEARNING IN CYBER

Module Number: COMP70049

Title of Assignment: Assignment

Contribution to mark: This assignment is worth 100% of the overall mark for this module.

Hand in deadlines: **Refer to the Blackboard**

Introduction

This assignment explores the intersection of **machine learning** and **cybersecurity**, focusing on key areas such as **phishing email detection**, **network intrusion classification**, **anomaly detection**, and **ransomware identification**. Students will implement **both classic machine learning models** (e.g., **Logistic Regression**, **Decision Trees**, **Random Forests**, **Isolation Forests**, **SVMs**) and **deep learning approaches** (e.g., **LSTMs**, **CNNs**, **Autoencoders**, **Transformers**) to tackle real-world security challenges.

The primary objectives of this assignment are:

- To **understand the application of machine learning** in cybersecurity.
- To **develop hands-on skills** in data preprocessing, feature engineering, and model training.
- To **compare classic ML vs. deep learning models** in different cybersecurity use cases.
- To **evaluate the performance of ML models** using relevant metrics such as **accuracy**, **precision**, **recall**, and **F1-score**.

The assignment is divided into **four sections**, each focusing on a different aspect of cybersecurity:

1. **Email Security & Phishing Detection (25 Marks)** – Detecting phishing emails using text-based ML and deep learning models.
2. **Cyber Attack Detection (25 Marks)** – Classifying different types of network intrusions using structured attack datasets.
3. **Anomaly Detection for Cyber Threats (25 Marks)** – Identifying unusual or suspicious network activity using unsupervised ML techniques.

4. **Ransomware Detection & Prevention (25 Marks)** – Detecting ransomware activity based on system behavior and API call sequences.

Each section carries 25 marks, making a total of 100 marks for the assignment.

Instructions

1. Implementation Requirements

For each of the four sections in this assignment, students must:

1. **Choose an appropriate dataset** (introduced in the assignment or sourced online).
2. **Preprocess the data**, including cleaning, feature engineering, and normalization.
3. **Implement two machine learning models** for each section: **One classic machine learning algorithm** (e.g., Logistic Regression, Random Forest, SVM, Decision Tree) and **One deep learning model** (e.g., LSTM, CNN, Autoencoder, BERT).
4. **Train and evaluate the models** using appropriate performance metrics.
5. **Compare the results** of both models and analyze their effectiveness.
6. **Provide visualizations** (e.g., confusion matrix, ROC curve, loss/accuracy plots).

The implementation should be done in **Python** using appropriate libraries.

Each student must submit:

1. **Python script.** The code should be well-structured and commented. Ensure the notebook/script runs correctly without errors.

2. **Written Report** (up to 3000 words)

o **Include in the report for each experiment**

- Description of datasets used.
- Explanation of preprocessing techniques.
- Overview of implemented ML and deep learning models.
- Performance comparison of models.
- Visualizations (graphs, confusion matrices, accuracy/loss plots).
- Discussion on findings and cybersecurity implications.
- Summary of key takeaways and potential improvements.

3. A file containing the implemented code(s)

4. **README File**

- o Instructions on how to run the code.

Upload to the Blackboard via provided link by submission deadline.

Marking Criteria

Criteria	Description	Marks
Correctness of Implementation	Proper execution of ML and deep learning models	50
Depth of Analysis	Discussion of results, insights, and performance comparison	30
Code Quality & Documentation	Readability, comments, structure, and efficiency	10

Criteria	Description	Marks
Report Presentation	Clarity, organization, and completeness	10

Section 1: Email Security & Phishing Detection (25 Marks)

1. Task:

Phishing emails are a major cybersecurity threat. The goal is to build a model that can detect **phishing emails** based on textual features.

2. Steps to Follow:

1. Dataset Selection: Use datasets such as:

- o Enron Spam Dataset
- o SpamAssassin Dataset
- o Phishing Email Dataset from UCI Machine Learning Repository

2. Preprocessing:

- o Convert email text into numerical features using **TF-IDF** or **word embeddings (Word2Vec, GloVe)**.
- o Remove stopwords, punctuation, and apply stemming/lemmatization.

3. Classic Machine Learning Approach:

- o Train a **Logistic Regression** or **Random Forest** model for classification.

4. Deep Learning Approach:

- o Use an **LSTM (Long Short-Term Memory)** or **BERT (Bidirectional Encoder Representations from Transformers)** model for email classification.

5. Evaluation:

- o Compare models using **Precision, Recall, F1-score, and ROC curve**.

Section 2: Cyber Attack Detection (25 Marks)

1. Task:

Network intrusion detection is critical for identifying cyber-attacks. In this section, students will develop a machine learning model to classify various attack types.

2. Steps to Follow:

3. Dataset Selection: Use datasets such as:

- o **NSL-KDD Dataset**
- o **CIC-IDS2017 (Canadian Institute for Cybersecurity)**

4. Preprocessing:

- o Handle missing values, standardize numerical data, and encode categorical variables.
- o Perform **feature selection** using techniques like **PCA (Principal Component Analysis)**.

5. Classic Machine Learning Approach:

- o Train a **Decision Tree** or **Random Forest** model for multi-class attack classification.

6. Deep Learning Approach:

- o Use a **CNN (Convolutional Neural Network)** or **RNN (Recurrent Neural Network)** to process network traffic data.

7. Evaluation:

- o Compare model accuracy, precision-recall curves, and **Confusion Matrix**.

Section 3: Anomaly Detection for Cyber Threats (25 Marks)

1. Task:

Anomaly detection helps detect **unknown cyber threats** and **zero-day attacks** by identifying unusual network behavior. This section focuses on unsupervised learning techniques for cybersecurity.

2. Steps to Follow:

1. Dataset Selection: Use datasets such as:

- o **UNSW-NB15 Dataset**
- o **CIC-IDS2018 (Anomaly & Intrusion Detection Dataset)**

2. Preprocessing:

- o Remove duplicates, normalize numerical features, and handle missing values.
- o Select **relevant network traffic features** (e.g., packet size, protocol type, connection duration).

3. Classic Machine Learning Approach:

- o Train an **Isolation Forest** or **One-Class SVM** to detect anomalous behavior.

4. Deep Learning Approach:

- o Use an **Autoencoder Neural Network** to learn normal traffic patterns and identify anomalies.

5. Evaluation:

- o Measure **True Positive Rate (TPR)**, **False Positive Rate (FPR)**, and **Precision-Recall curves**.

Section 4: Ransomware Detection & Prevention (25 Marks)

1. Task:

Ransomware is a significant cybersecurity threat. This section focuses on detecting ransomware activity based on system behavior.

2. Steps to Follow:

1. Dataset Selection: Use datasets such as:

- o **Windows API Call Sequence Dataset** (from ransomware attacks)
- o **CIC-AndMal2017 (for Android Ransomware Detection)**

2. Preprocessing:

- o Convert system logs (API calls, file access patterns) into numerical feature representations.
- o Perform **feature extraction** to identify key ransomware behavior patterns.

3. Classic Machine Learning Approach:

- o Train an **SVM (Support Vector Machine)** or **Gradient Boosting Classifier** to classify ransomware vs. normal activity.

4. Deep Learning Approach:

- o Use **LSTM (Recurrent Neural Network)** to model system behavior over time and detect ransomware attacks.

5. **Evaluation:**

- o Compare models using **Precision, Recall, and F1-score**.